Bhushan Ladgaonkar

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Professional Summary

Security Researcher and Machine Learning Engineer specializing in **Adversarial AI** and **AI Security**. Published first-author researcher in adversarial robustness; proven experience engineering ML-based Network Intrusion Detection Systems for enterprise infrastructure and building hardware-optimized local LLM inference systems on Apple Silicon. Seeking to leverage expertise in adversarial robustness, simulation, and security architectures.

Education

Fr. Conceicao Rodrigues College of Engineering — Mumbai, India
Bachelor of Technology (B.Tech) in Computer Engineering — *Expected May 2028*

Experience

NetFortress Technologies Private Limited — Remote
Cybersecurity Research Intern — *June 2026 – June 2026*

- Assisting in product development and technical research for enterprise technology solutions, focusing heavily on cybersecurity and AI infrastructure.
- Supporting AI-related and cybersecurity initiatives through active participation in testing, execution activities, and the preparation of comprehensive technical documentation and reports.

Oil and Natural Gas Corporation (ONGC) — Mumbai, India
Summer Intern – Database and Security Group — *May 2025 – June 2025*

- Developed a prototype **Network Intrusion Detection System (NIDS)** on the UNSW-NB15 dataset (2M+ records) to identify zero-day threats, addressing limitations of traditional signature-based detection.
- Implemented **Random Forest** and **XGBoost** classifiers with **SMOTE**-based class imbalance handling and advanced feature engineering, reducing false positive rates while sustaining high detection sensitivity.
- Collaborated with the Database Group to document threat detection methodologies, translating raw data findings into actionable security intelligence.

Publications

Benchmarking Resilience: Asymmetric Latent Purification Inspired by Generative Diffusion Bottlenecks
Published: May 2026

First & Corresponding Author · IJRIAS · doi.org/10.51584/IJRIAS.2026.110400185

- Proposed the **Asymmetric Latent Purifier (ALP)**: a zero-shot structural defense leveraging a 64-channel bottleneck to shatter adversarial gradients prior to classification, eliminating the need for full generative models.
- Recovered CIFAR-10 accuracy to **32.06%** under iterative **PGD-7** attacks (**25.68 dB PSNR**) while optimizing Apple Silicon inference to **1.25 ms/sample** (**100×+** faster than iterative purifiers).

Technical Projects

Gemma-M4-Agent: Local LLM Inference System | *Python, MLX, FastAPI, ChromaDB, React* — Apr 2026

- Built a full-stack local LLM agent around **Google Gemma** on Apple Silicon M4 via **MLX framework** – direct hardware-to-model ownership without wrappers such as Ollama or LM Studio; 8-bit quantized model footprint of **8.5 GB** on Unified Memory.
- Engineered a **Hybrid Memory Architecture** decoupling SSD-resident model weights from USB-persistent **ChromaDB** vector store (384-dim embeddings), enabling zero-copy GPU inference with persistent long-term memory across reboots.
- Implemented **dual-gate stop-token defense**: silicon-level halt via MLX kernel token IDs and software-level control-sequence buffer matching, achieving **100% stop-token accuracy** and preventing hallucinated output beyond semantic boundaries.

Headless Kali: Zero-Trust Remote Workstation | *WireGuard, OpenSSH, XRDP, Linux ACPI, LUKS* — Oct 2026

- Architected a zero-trust remote workstation for an encrypted (**LUKS**) Kali Linux laptop using a **WireGuard P2P overlay** with STUN hole-punching and **OpenSSH loopback forwarding** of XRDP across CGNAT networks, achieving a direct **9 ms RTT** link with **zero open WAN/LAN ports** (verified via Nmap).
- Hardened Linux ACPI power states (masked systemd sleep targets, forced Wi-Fi D0 state) to sustain **200+ hours of continuous uptime** with the lid closed; replaced GNOME with a headless **XRDP/XFCE** session, cutting idle RAM by **∼65%** (1.4 GB to 450 MB) and stream bandwidth by **∼70%** (to ∼1.5 Mbps).

Technical Skills

Languages: Python, C, SQL, Java
AI and ML: Machine Learning, Reinforcement Learning, RAG, LLMs, MCTS, PyTorch, MLX, ChromaDB, XGBoost
Security: Network Intrusion Detection (NIDS), Adversarial Robustness, OSINT, Reconnaissance, WireGuard, Linux (Kali, Ubuntu)
Cloud and Developer Tools: AWS, Google Cloud, Git, FastAPI, React, Node.js, Docker, VS Code, Jupyter Notebooks

Certifications

CS50 Cybersecurity – Harvard University
Comprehensive study of encryption, web security, network protocols, and defensive programming practices.

AWS Academy Cloud Architecting – Amazon Web Services
Designing scalable, resilient, and secure cloud architectures on AWS.

Google Cloud Engineering Certificate – Google Cloud
Foundational cloud engineering skills on Google Cloud, covering infrastructure, deployment, and operations.